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Substrate processing method and substrate processing apparatus

Abstract

A substrate processing method of etching an etching target film formed on a substrate includes: preparing the substrate having the etching target film; and etching the etching target film, wherein the etching the etching target film includes repeating, a plurality of times, supplying an etchant gas, and plasma-exciting a reaction gas to expose the substrate to plasma of the reaction gas.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a U.S. National Stage Entry of International Patent Application No. PCT/JP2021/007986, filed Mar. 2, 2021, which claims the benefit of priority to Japanese Patent Application No. 2020-042179, filed Mar. 11, 2020, each of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a substrate processing method and a substrate processing apparatus.

BACKGROUND

(3) For example, a substrate processing apparatus for etching a silicon-based film formed on a substrate has been known.

(4) Patent Document 1 discloses a method of etching a layer on a substrate in a semiconductor processing chamber, which includes: introducing a first gas, which is an etchant gas appropriate for etching the layer, into the chamber; retaining the first gas in the chamber for a certain period of

time sufficient to adsorb at least a part of the first gas in the layer; substantially replacing the first gas in the chamber with an inert gas; generating a metastable gas from the inert gas; and etching the layer with the metastable gas.

PRIOR ART DOCUMENT

Patent Document

(5) Patent Document 1: Japanese Patent Laid-open Publication No. 2014-522104.

SUMMARY

(6) In one aspect, the present disclosure provides a substrate processing method and a substrate processing apparatus capable of obtaining good etching characteristics.

(7) In order to solve the above-mentioned problem, according to one aspect, there is provided a substrate processing method of etching an etching target film formed on a substrate. The method includes: preparing the substrate having the etching target film; and etching the etching target film, wherein the etching the etching target film includes repeating, a plurality of times, supplying an etchant gas, and plasma-exciting a reaction gas to expose the substrate to plasma of the reaction gas.

(8) According to one aspect, it is possible to provide a substrate processing method and a substrate processing apparatus capable of obtaining good etching characteristics.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view showing a configuration example of a plasma processing apparatus.

(2) FIG. 2 is a flowchart illustrating an example of a substrate processing by the plasma processing apparatus.

(3) FIG. 3 is a time chart illustrating an example of an etching process by the plasma processing apparatus.

(4) FIG. 4 is an example of a graph illustrating a relationship between an etching temperature and an etching amount for each film type of a silicon-based film.

(5) FIG. 5 is an example of a graph illustrating a selectivity between an amorphous silicon film and another silicon-based film (SiN film, SiO₂ film).

(6) FIG. 6 is an example of graphs illustrating a relationship between a process temperature and an etching amount per cycle.

(7) FIG. 7 is an example of a graph illustrating an etching amount of a germanium-based film.

DETAILED DESCRIPTION

(8) Hereinafter, embodiments for carrying out the present disclosure will be described with reference to the drawings. In each drawing, the same components will be denoted by the same reference numerals, and redundant explanations thereof may be omitted.

(9) <Substrate Processing Apparatus>

(10) A plasma processing apparatus (substrate processing apparatus) **100** according to the present embodiment will be described with reference to FIG. 1. FIG. 1 is a schematic view showing a configuration example of a plasma processing apparatus **100**.

(11) The plasma processing apparatus **100** includes a cylindrical processing container **1** with a ceiling and an open lower end. The entire processing container **1** is made of, for example, quartz. A ceiling plate **2** made of quartz is provided in the processing container **1** in a vicinity of an upper end of the processing container **1**, and a region below the ceiling plate **2** is sealed. A metallic manifold **3** molded into a cylindrical shape is connected to an opening at the lower end of the processing container **1** via a seal **4** such as an O-ring.

(12) The manifold **3** supports the lower end of the processing container **1**, and a wafer boat **5**, in which a plurality of (e.g., 25 to 150) semiconductor wafers (hereinafter referred to as “substrates

W”) is loaded in multiple stages, is inserted into the processing container **1** from below the manifold **3**. The plurality of substrates **W** is accommodated substantially horizontally in the processing container **1** at intervals along a vertical direction. The wafer boat **5** is made of, for example, quartz. The wafer boat **5** has three rods **6** (two of which are shown in FIG. **1**), and the plurality of substrates **W** is supported by grooves (not shown) formed in the rods **6**.

(13) The wafer boat **5** is placed on a table **8** with a thermal insulating tube **7** interposed therebetween. The thermal insulating tube **7** is made of quartz. The table **8** is supported on a rotary shaft **10**, which penetrates a metallic (stainless steel) lid **9** configured to open and close an opening at a lower end of the manifold **3**.

(14) A magnetic fluid seal **11** is provided at the penetrating portion of the rotary shaft **10** to airtightly seal and rotatably support the rotary shaft **10**. A seal **12** for maintaining airtightness inside the processing container **1** is provided between a peripheral portion of the lid **9** and the lower end of the manifold **3**.

(15) The rotary shaft **10** is provided at a tip of an arm **13** supported by an elevating mechanism (not shown) such as a boat elevator. The wafer boat **5** and the lid **9** are integrally moved up and down to be inserted and separated with respect to an interior of the processing container **1**. In addition, the table **8** may be fixedly provided on the lid **9** to process the substrates **W** without rotating the wafer boat **5**.

(16) In addition, the plasma processing apparatus **100** includes a gas supply **20** that supplies predetermined gases such as a processing gas and a purge gas into the processing container **1**.

(17) The gas supply **20** includes gas pipes **21**, **22**, and **24**. The gas pipe **21** is made of, for example, quartz. The gas pipe **21** is a quartz pipe inwardly penetrating a sidewall of the manifold **3** and then being bent upward. The gas pipe **22** is made of, for example, quartz. The gas pipe **22** inwardly penetrates the sidewall of the manifold **3**, is bent upward, and then extends vertically. A plurality of gas holes **22g** is formed at predetermined intervals in a vertical portion of the gas pipe **22** over a vertical length corresponding to a wafer support range of the wafer boat **5**. Each gas hole **22g** discharges a gas in a horizontal direction. The gas pipe **24** is made of, for example, quartz. The gas pipe **24** is a short quartz pipe penetrating the sidewall of the manifold **3**.

(18) An etchant gas is supplied from an etchant gas source **21a** to the gas pipe **21** via a gas line. The gas line is provided with a flow rate controller **21b** and an on-off valve **21c**. Thus, the etchant gas from the etchant gas source **21a** is supplied into the processing container **1** via the gas line and the gas pipe **21**. As the etchant gas, hydrogen fluoride (HF), for example, may be used. In addition, the etchant gas is not limited thereto, and a hydrogen halide and a halogenated compound such as F.sub.2, Cl.sub.2, Br.sub.2, I.sub.2, HCl, BCl.sub.3, HBr, HI, NF.sub.3, ClF.sub.3, CF.sub.4, and the like may be used.

(19) The vertical portion (vertically-extending portion in which the gas holes **22g** are formed) of the gas pipe **22** is provided in a plasma generation space described below. A reaction gas containing hydrogen is supplied from a reaction gas source **22a** to the gas pipe **22** via a gas line. The gas line is provided with a flow rate controller **22b** and an on-off valve **22c**. In addition, nitrogen gas (N.sub.2) is supplied from a reaction gas source **23a** to the gas pipe **22** via a gas line. The gas line is provided with a flow rate controller **23b** and an on-off valve **23c**. Thus, a gas mixture of the reaction gas containing hydrogen and the nitrogen gas from the reaction gas sources **22a** and **23a** is supplied to the plasma generation space via the gas lines and the gas pipe **22**, and is plasmarized in the plasma generation space and supplied into the processing container **1**. As the reaction gas containing hydrogen, NH.sub.3 gas or H.sub.2 gas, for example, may be used. In addition, the reaction gas is not limited thereto, and a gas containing at least hydrogen (H) or deuterium (D) such as H.sub.2, N.sub.2H.sub.4, C.sub.2H.sub.4, NH.sub.3, or D.sub.2 may be used.

(20) A purge gas is supplied from a purge gas source (not shown) to the gas pipe **24** via a gas line. The gas line (not shown) is provided with a flow rate controller (not shown) and an on-off valve (not shown). Thus, the purge gas from the purge gas source is supplied into the processing

container **1** via the gas line and the gas pipe **24**. As the purge gas, an inert gas such as argon (Ar) or nitrogen (N.sub.2), for example, may be used. Although the case where the purge gas is supplied from the purge gas source into the processing container **1** via the gas line and the gas pipe **24** has been described, the present disclosure is not limited thereto. The purge gas may be supplied from any of the gas pipes **21** and **22**.

(21) A plasma generation mechanism **30** is formed on a part of the sidewall of the processing container **1**. The plasma generation mechanism **30** plasmarizes the NH.sub.3 gas (or H.sub.2 gas) to generate hydrogen (H) radicals, and plasmarizes the N.sub.2 gas to generate active species for nitration.

(22) The plasma generation mechanism **30** includes a plasma partition wall **32**, a pair of plasma electrodes **33** (one of which is shown in FIG. **1**), a feed line **34**, a radio frequency power supply **35**, and an insulating protective cover **36**.

(23) The plasma partition wall **32** is airtightly welded to an outer wall of the processing container **1**. The plasma partition wall **32** is made of, for example, quartz. The plasma partition wall **32** has a concave cross section, and covers an opening **31** formed in the sidewall of the processing container **1**. The opening **31** is vertically elongated so as to cover all the substrates **W** supported in the wafer boat **5** in the vertical direction. The gas pipe **22** for discharging the gas mixture of the NH.sub.3 gas and the N.sub.2 gas is disposed in an inner space, which is defined by the plasma partition wall **32** and in communication with the interior of the processing container **1**, i.e., the plasma generation space. In addition, the gas pipe **21** for discharging the etchant gas is provided at a position close to the substrates **W** along the inner sidewall of the processing container **1** outside the plasma generation space.

(24) Each of the pair of plasma electrodes **33** (one of which is shown in FIG. **1**) has an elongated shape. The plasma electrodes **33** are disposed on opposite sides of an outer wall surface of the plasma partition wall **32** so as to face each other along the vertical direction. Each plasma electrode **33** is held by, for example, a holder (not shown) provided on a side surface of the plasma partition wall **32**. The feed line **34** is connected to a lower end of each plasma electrode **33**.

(25) The feed line **34** electrically connects each plasma electrode **33** to the radio frequency power supply **35**. In the shown example, the feed line **34** has one end connected to the lower end of each plasma electrode **33** and the other end connected to the radio frequency power supply **35**.

(26) The radio frequency power supply **35** is connected to the lower end of each plasma electrode **33** via the feed line **34**, and supplies radio frequency power of, for example, 13.56 MHz to the pair of plasma electrodes **33**. Thus, the radio frequency power is applied to the plasma generation space defined by the plasma partition wall **32**. The NH.sub.3 gas (or H.sub.2 gas) discharged from the gas pipe **22** is plasmarized in the plasma generation space to which the radio frequency power has been applied, and hydrogen radicals thus generated are supplied into the processing container **1** via the opening **31**. In addition, the N.sub.2 gas supplied from the gas pipe **22** is plasmarized in the plasma generation space to which the radio frequency power has been applied, and active species for nitration thus generated are supplied into the processing container **1** via the opening **31**.

(27) The insulating protective cover **36** is provided outside the plasma partition wall **32** so as to cover the plasma partition wall **32**. A coolant passage (not shown) is provided in an inner portion of the insulating protective cover **36**, and the plasma electrodes **33** are cooled by flowing a coolant such as a cooled nitrogen (N.sub.2) gas through the coolant passage. Further, a shield (not shown) may be provided between the plasma electrodes **33** and the insulating protective cover **36** so as to cover the plasma electrodes **33**. The shield is made of, for example, a good conductor such as a metal, and is grounded.

(28) An exhaust port **40** for vacuum evacuation of the processing container **1** is provided in a sidewall portion of the processing container **1** facing the opening **31**. The exhaust port **40** is vertically elongated so as to correspond to the wafer boat **5**. An exhaust port cover **41**, which is molded into a U-shaped cross section, is attached to a portion of the processing container **1**

corresponding to the exhaust port **40** so as to cover the exhaust port **40**. The exhaust port cover **41** extends upward along the sidewall of the processing container **1**. An exhaust pipe **42** for evacuating the processing container **1** via the exhaust port **40** is connected to a lower portion of the exhaust port cover **41**. The exhaust pipe **42** is connected to an exhaust device **44**, which includes a pressure control valve **43** for controlling a pressure in the processing container **1**, a vacuum pump, and the like. The interior of the processing container **1** is evacuated by the exhaust device **44** via the exhaust pipe **42**.

(29) In addition, a cylindrical heating mechanism **50** for heating the processing container **1** and the substrates **W** inside the processing container **1** is provided so as to surround an outer periphery of the processing container **1**.

(30) In addition, the plasma processing apparatus **100** includes a controller **60**. The controller **60** controls, for example, operations of respective components of the plasma processing apparatus **100**, such as supply and stop of respective gases by opening and closing the on-off valves **21c** to **23c**, gas flow rates by the flow rate controllers **21b** to **23b**, and evacuation by the exhaust device **44**. In addition, the controller **60** controls, for example, on and off of the radio frequency power by the radio frequency power supply **35** and the temperature of the substrates **W** by the heating mechanism **50**.

(31) The controller **60** may be, for example, a computer or the like. In addition, a computer program that operates respective components of the plasma processing apparatus **100** is stored in a storage medium. The storage medium may be, for example, a flexible disk, a compact disk, a hard disk, a flash memory, a DVD, or the like.

(32) Next, an example of a substrate processing by the plasma processing apparatus **100** shown in FIG. **1** will be described. FIG. **2** is a flowchart illustrating an example of a substrate processing by the plasma processing apparatus **100**. The plasma processing apparatus **100** etches a silicon-based film as an etching target film formed on the substrate **W**.

(33) In step **S101**, the substrate **W** having a silicon-based film is prepared. Specifically, the substrate **W** having the silicon-based film is set in the wafer boat **5**. The arm **13** inserts the wafer boat **5** into the processing container **1** from the lower end of the manifold **3**. Thereafter, the processing container **1** is made airtight by the lid **9**.

(34) In step **S102**, the silicon-based film of the substrate **W** is etched.

(35) An etching process by the plasma processing apparatus **100** will be described with reference to FIG. **3**. FIG. **3** is a time chart illustrating an example of an etching process by the plasma processing apparatus **100**.

(36) The etching process shown in FIG. **3** is a process of repeating a cycle a predetermined number of times, wherein the cycle includes step **S201** of supplying an etchant gas, step **S202** of purging, step **S203** of applying RF power and supplying a reaction gas, and step **S204** of purging, and alternately supplies the etchant gas and the reaction gas to etch a silicon-based film formed on the substrate **W**. FIG. **3** shows only one cycle. In addition, in steps **S201** to **S204**, N.sub.2 gas as a purge gas is continuously supplied from the gas pipe **24** during the etching process.

(37) Step **S201** of supplying the etchant gas is a process of supplying the etchant gas into the processing container **1**. In step **S201** of supplying the etchant gas, first, by opening the on-off valve **21c**, the etchant gas is supplied from the etchant gas source **21a** into the processing container **1** via the gas pipe **21**. Thus, the etchant gas (HF gas) fluorinates a surface of the silicon-based film of the substrate **W**, such that the surface of the silicon-based film is brought into a state in which the fluorination reaction is saturated.

(38) Step **S202** of purging is a process of purging an excess etchant gas or the like in the processing container **1**. In step **S202** of purging, the on-off valve **21c** is closed to stop the supply of the etchant gas. Thus, a purge gas, which is continuously supplied from the gas pipe **24**, purges the excess etchant gas or the like in the processing container **1**.

(39) Step **S203** of supplying the reaction gas is a process of supplying a gas mixture of NH.sub.3

gas and N.sub.2 gas as the reaction gas. In step S203 of supplying the reaction gas, by opening the on-off valves 22c and 23c, the reaction gas is supplied from the reaction gas sources 22a and 23a into the plasma partition wall 32 via the gas pipe 22. In addition, the RF power is applied to the plasma electrodes 33 by the radio frequency power supply 35 to generate plasma inside the plasma partition wall 32. Hydrogen (H) radicals and active species for nitration are generated and supplied into the processing container 1 from the opening 31. Thus, the hydrogen (H) radicals and the active species for nitration react with the surface of the fluorinated silicon-based film, such that a surface layer (one layer) of the fluorinated silicon-based film is etched.

(40) Specifically, the hydrogen (H) radicals and the active species for nitration are supplied to the surface of the fluorinated silicon-based film, and a reaction product including silicon and fluorine such as (NH.sub.4).sub.2SiF.sub.6 is obtained. The generated (NH.sub.4).sub.2SiF.sub.6 is sublimated and discharged from the interior of the processing container 1 by the exhaust device 44. Thus, the surface of the silicon-based film is etched. The reaction product is a composition including Si and F, and is volatilized and removed in a temperature and pressure range of this process.

(41) Step S204 of purging is a process of purging an excess reaction gas or reaction product (NH.sub.4).sub.2SiF.sub.6, or the like in the processing container 1. In step S204 of purging, the on-off valves 22c and 23c are closed to stop the supply of the reaction gas. Thus, the purge gas continuously supplied from the gas pipe 24 purges the excess reaction gas or reaction product, or the like inside the processing container 1.

(42) By repeating the cycle described above, the silicon-based film formed on the substrate W is etched.

(43) In a plasma etching process by the plasma processing apparatus 100 according to the present embodiment, a silicon-based film is etched by repeating fluorination of a film surface by an etchant gas, generation of a reaction product by reaction of the fluorinated film surface with plasma of a reaction gas, and sublimation of the generated reaction product from the substrate W. Thus, it is possible to uniformly (isotropically) etch the silicon-based film. That is, even a silicon-based film having a high aspect ratio recess may be uniformly etched on an entry side and an inner side of a sidewall of the recess. In addition, after performing the fluorination uniformly, by controlling activity of the reaction gas by the plasma or drawing of the active species along a depth direction, it is also possible to perform a control to preferentially etch an opening side of the recess, or the top and bottom of the recess.

(44) Here, desirable ranges of a condition for etching the silicon-based film using the etchant gas and the reaction gas in step S102 are shown below: Temperature: 25 degrees C. to 400 degrees C.; Pressure: 0.1 Torr to 50.0 Torr; HF gas flow rate: 100 sccm to 5,000 sccm; NH.sub.3 gas flow rate: 1,000 sccm to 10,000 sccm; N.sub.2 gas flow rate: 50 sccm to 10,000 sccm; step S201 time: 5.0 seconds to 180 seconds; step S202 time: 5.0 seconds to 60 seconds; step S203 time: 5.0 seconds to 300 seconds; step S204 time: 5.0 seconds to 60 seconds; and RF power: 50 W to 500 W.

(45) FIG. 4 is an example of a graph showing a relationship between an etching temperature and an etching amount for each film type of a silicon-based film. The horizontal axis indicates the etching temperature (degrees C.), and the vertical axis indicates the etching amount (A). In addition, a result when the etching target is an amorphous silicon film is indicated by the solid line, a result when the etching target is a SiN film is indicated by the dashed line, and a result when the etching target is a SiO.sub.2 film is indicated by the one-dot dashed line.

(46) FIG. 5 is an example of a graph showing an etching selectivity between an amorphous silicon film and another silicon-based film (SiN film, SiO.sub.2 film). The horizontal axis indicates the etching temperature (degrees C.), and the vertical axis indicates the selectivity. In addition, a selectivity of the amorphous silicon film to the SiN film is indicated by the dashed line, and a selectivity of the amorphous silicon film to the SiO.sub.2 film is indicated by the one-dot dashed line.

(47) As shown in FIGS. 4 and 5, with the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, it is possible to appropriately etch the amorphous silicon film. Further, with the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, it is possible to selectively etch the amorphous silicon film with respect to the SiN film or the SiO.sub.2 film.

(48) In addition, in the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, a silicon film, a polysilicon film, or a crystalline silicon film also has high selectivity with respect to the SiN film or the SiO.sub.2 film. Therefore, with the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, it is possible to selectively etch the silicon film, the polysilicon film, or the crystalline silicon with respect to the SiN film or the SiO.sub.2 film.

(49) FIG. 6 is an example of graphs showing a relationship between a process temperature and an etching amount per cycle (EPC). The horizontal axis indicates the process temperature (degrees C.), and the vertical axis indicates the etching amount per cycle (A/cycle).

(50) In graphs **301** to **304**, the supply pressure and the supply time of the HF gas in step **201** (see FIG. 3) were set different from one another. Specifically, in graph **301**, they were set to 1 Torr and 10 seconds. In graph **302**, they were set to 5 Torr and 30 seconds. In graph **303**, they were set to 9 Torr and 60 seconds. In graph **304**, they were set to 27 Torr and 60 seconds.

(51) In addition, the supply flow rate of the HF gas was set to 1 slm in common in graphs **301** to **304**. The supply flow rate, the supply pressure, and the supply time of the NH.sub.3 gas in step **203** (see FIG. 3) were set to 5 slm, 0.3 Torr, and 10 seconds, respectively, in common in graphs **301** to **304**.

(52) As shown in FIG. 6, in the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, by controlling the supply pressure and the supply time of the HF gas, etching can be performed within a wide temperature range.

(53) As described above, in the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, it is possible to determine the etching temperature based on the supply pressure and the supply time of the HF gas. In other words, good etching characteristics can be obtained by appropriately selecting the supply pressure and the supply time of the HF gas with respect to a desired etching temperature.

(54) In addition, in the vicinity of 400 degrees C., as shown in region **305**, an increase in the etching amount per cycle was observed. The reason is because thermal etching components were superimposed due to an increase in thermal activity of HF.

(55) In addition, in the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, the etching target film may be a germanium-based film. Here, the germanium-based film includes at least one film selected from the group consisting of an amorphous silicon germanium film, a polycrystalline silicon germanium film, a single crystalline silicon germanium film, an amorphous germanium film, a polycrystalline germanium film, and a single crystalline germanium film.

(56) FIG. 7 is an example of a graph showing an etching amount of a germanium-based film. The horizontal axis indicates a ratio of germanium (Ge) in a silicon-germanium-based film (SiGe), and the vertical axis indicates an etching amount (A). Further, a result when the process temperature is 75 degrees C. is indicated by the solid line, and a result when the process temperature is 100 degrees C. is indicated by the dashed line.

(57) Here, as shown in comparison with FIGS. 4 and 7, the germanium-based film (see FIG. 7) has a higher etching amount than the SiN film or the SiO.sub.2 film (see FIG. 4). That is, in the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, the germanium-based film also has high selectivity with respect to the SiN film or the SiO.sub.2 film. Therefore, with the plasma etching process by the plasma processing apparatus **100** according to the present embodiment, it is possible to selectively etch the germanium-based film with respect to

the SiN film or the SiO.sub.2 film.

(58) Although the etching method of the present embodiment by the plasma processing apparatus **100** has been described above, the present disclosure is not limited to the above-described embodiment and the like, and various modifications and improvements can be made within the scope of the gist of the present disclosure described in the claims.

(59) In step **S101**, the process of preparing the substrate W having a silicon-based film (etching target film) has been described as setting the substrate W having the silicon-based film in the processing container **1**, but is not limited thereto. It may be a process of forming a silicon-based film on the substrate W in the processing container **1** by the plasma processing apparatus **100**.

(60) This application claims priority based on Japanese Patent Application No. 2020-42179 filed on Mar. 11, 2020, and the entire contents of this Japanese patent application is incorporated herein by reference.

EXPLANATION OF REFERENCE NUMERALS

(61) **W**: substrate, **100**: plasma processing apparatus (substrate processing apparatus), **1**: processing container, **2**: ceiling plate, **20**: gas supply, **21**, **22**, **24**: gas pipe, **21**: gas pipe, **22**: gas pipe, **22g**: gas hole, **24**: gas pipe, **21a**: etchant gas source, **22a**, **23a**: reaction gas source, **30**: plasma generation mechanism (radio frequency power supply), **44**: exhaust device, **50**: heating mechanism, **60**: controller

Claims

1. A substrate processing method of etching an etching target film formed on a substrate, the method comprising: preparing the substrate having the etching target film in a processing container; and etching the etching target film in the processing container, wherein the etching the etching target film includes repeating, a plurality of times: supplying an etchant gas; and plasma-exciting a reaction gas to expose the substrate to plasma of the reaction gas, and wherein the supplying the etchant gas includes controlling a pressure in the processing container and a supply time of the etchant gas according to an etching temperature of the etching target film.
 2. The substrate processing method of claim 1, wherein the etchant gas includes a hydrogen halide.
 3. The substrate processing method of claim 2, wherein the reaction gas includes H.
 4. The substrate processing method of claim 3, wherein the reaction gas is a gas mixture of ammonia gas or hydrogen gas and nitrogen gas.
 5. The substrate processing method of claim 4, wherein the etching target film includes a silicon-based film, and includes at least one selected from the group consisting of a silicon film, an amorphous silicon film, a polysilicon film, and a crystalline silicon film.
 6. The substrate processing method of claim 5, wherein the etching target film includes a germanium-based film, and includes at least one selected from the group consisting of an amorphous silicon germanium film, a polycrystalline silicon germanium film, a single crystalline silicon germanium film, an amorphous germanium film, a polycrystalline germanium film, and a single crystalline germanium film.
 7. The substrate processing method of claim 1, wherein the reaction gas includes H.
 8. The substrate processing method of claim 1, wherein the etching target film includes a silicon-based film, and includes at least one selected from the group consisting of a silicon film, an amorphous silicon film, a polysilicon film, and a crystalline silicon film.
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